Further results on the brief *Physical vector spaces for quantum mechanics*

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Abstract

This note collects results that go beyond the technical brief *Physical vector spaces for quantum mechanics* [1] and partial answers to its open problems. They are not part of the brief. Each section is self-contained, and each names the Lean declarations that check it; the Lean files are in the brief's directory, under `QuantumVectorSpaces/FurtherResults/`.

1 The Hilbert brief's change of frame is not a map of Schwartz space

The brief proposes quantum vector spaces (QSTO:V) as a replacement for the Hilbert-space formulation QSTO:H, and argues in the proof of its Theorem 5 that QSTO:V satisfies [B]WDQ. The argument assumes that "the defining observables of A must be observables in B and vice-versa" for two frames A and B of the same system. This premise holds exactly when the two frames give the same topology on V (checked with the brief, `QuantumVectorSpaces.topology_eq_iff_continuous`). It is not automatic. This section shows that it fails for the brief's own Example 3, $V = \mathcal{S}(\mathbb{R})$, with the change of frame that the companion brief *Unphysicality of Hilbert spaces for quantum mechanics* [2] uses against QSTO:H.

That change of frame is $y = \tan(\frac{\pi}{2}\operatorname{erf} x)$. Its unitary U on $L^2(\mathbb{R})$ sends the Gaussian packet

$$\psi(x) = \left(e^{-x^2}/\sqrt{\pi}\right)^{1/2} = \pi^{-1/4}e^{-x^2/2} \quad \text{to} \quad \phi(y) = \left(\pi(1+y^2)\right)^{-1/2}.$$

Theorem 1.1. *1. ψ is a Schwartz function, and $U\psi = \phi$ is not. So U does not map $\mathcal{S}(\mathbb{R})$ into itself.*

2. The position Y of the second frame, carried back to the first as $U^{-1}YU$, is not defined at the Schwartz function ψ.

Proof. Every Schwartz function f lies in the domain of the position operator, since xf is again a Schwartz function and so in L^2. But $\int y^2|\phi(y)|^2\,dy = \int y^2/(\pi(1+y^2))\,dy = \infty$, so ϕ is not in that domain (the brief *Unphysicality of Hilbert spaces*, Proposition 3). This gives (1). For (2), $\psi \in D(U^{-1}YU)$ iff $U\psi \in D(Y)$, which fails. □

So the two observers do not describe the same V with two families of defining observables, as the premise requires: the second observer cannot express the first observer's ψ in $\mathcal{S}(\mathbb{R})$. QSTO:V gives [B]WDQ only for changes of frame whose unitaries map V onto itself. The restriction is stated in print by Carcassi, Calderón and Aidala: on the Schwartz space "the only unitary transformations we are interested in are those that map Schwartz functions to Schwartz functions", and the change

of coordinate "is automatically ruled out" [3, Appendix, Sec. VIII]. The brief does not repeat it. Theorem 1.1 is the concrete instance, checked in Lean.

Which changes of variable do map $\mathcal{S}(\mathbb{R})$ into itself is known: Galbis and Jordá characterize them [4, Thm. 2.3]. That the restriction cannot be lifted by choosing a different space is shown in the companion note on the Hilbert brief [20, Sec. 2]: no nonzero state of $L^2(\mathbb{R})$ has finite position variance in every smooth frame both now and after a free evolution. A quantum vector space has to come with its group of changes of frame (Sec. 3).

1.1 Lean

Module:
`QuantumVectorSpaces.FurtherResults.SchwartzFrame`,
namespace `QuantumVectorSpaces.FurtherResults`. The unitary U (`frameUnitary`), the packets and the position operator are those of the Hilbert brief's formalization.

- `toLp_gaussSchwartz`: ψ is $\pi^{-1/4}$ times TauCeti's Gaussian Schwartz map.
- `toLp_mem_positionDomain`: every Schwartz function lies in the domain of the position operator on $L^2(\mathbb{R})$.
- `frameUnitary_gaussState_not_mem_range`, `not_forall_frameUnitary_mem_range`: Theorem 1.1(1).
- `toLp_gaussSchwartz_not_mem_domain`: Theorem 1.1(2).

The statement is made in $L^2(\mathbb{R})$, where U lives; it does not need U to be smooth.

2 The completeness problem of Definition 1

The brief leaves one problem with its Definition 1 open (Sec. 5, "Completeness of the Lie algebra"). The definition admits two cases that the brief calls pathological: "a Hilbert space with the defining observables of a qubit", which has "too many vectors", and "a Schwartz space with only position in the defining observable", which has "too few observables". This section shows that the second case is not a quantum vector space at all, and that the first one is excluded by an irreducibility condition which Examples 3 and 4 of the brief satisfy.

Throughout, V carries an inner product and defining observables X_i, and $\mathcal{A}$ is the unital algebra they generate. The topology of the brief's Corollary 2 is given by the seminorms $p_A(v) = \|Av\|$, $A \in \mathcal{A}$; Definition 1 requires V to be complete and second countable in it.

2.1 Position alone

Theorem 2.1. *Let $V = \mathcal{S}(\mathbb{R})$ with the L^2 inner product, and let position X be the only defining observable. Then V is not complete in the topology of Corollary 2. So it is not a quantum vector space, whatever $\hbar$ and structure constants are chosen.*

Proof. Let $\psi_0(x) = \pi^{-1/4} e^{-x^2/2}$ and $f_m = u_m \psi_0$ with $u_m(x) = \sqrt{(m+1)^{-2} + x^2}$, a Schwartz function since u_m is smooth with polynomially bounded derivatives. As $u_m(x) = |(m+1)^{-1} + ix|$, the triangle inequality gives $|u_m - u_k| \le |(m+1)^{-1} - (k+1)^{-1}|$ and $|u_m(x) - |x|| \le (m+1)^{-1}$. Every $A \in \mathcal{A}$ commutes with multiplication by u_m, so

$$p_A(f_m - f_k) = \|(u_m - u_k)\, A\psi_0\| \le \left|(m+1)^{-1} - (k+1)^{-1}\right| \|A\psi_0\|,$$

and (f_m) is a Cauchy sequence. If it converged to some $h \in \mathcal{S}(\mathbb{R})$, then $\|f_m - h\| \to 0$ (take $A = 1$), while $\|f_m - |x|\psi_0\| \leq \|\psi_0\|/(m+1)$. So $h = |x|\psi_0$ almost everywhere, and everywhere since both are continuous. But $|x|\psi_0$ is not differentiable at 0. □

So the brief's second example is excluded by Definition 1 already, through completeness. The pathology it was meant to show does not go away, though. The completion of V is the space Φ_X of all $f \in L^2(\mathbb{R})$ with $x^k f \in L^2(\mathbb{R})$ for every k, and Φ_X with position alone is a quantum vector space: it is a separable Fréchet space, the projective limit of the weighted spaces $L^2((1+x^2)^k\,dx)$. Neither statement is formalized. It still has "too few observables": multiplication by e^{ix} commutes with position, on Φ_X as on $\mathcal{S}(\mathbb{R})$.

2.2 Irreducibility

Definition 2.2. V with the defining observables X_i is *irreducible* if every linear map $T\colon V \to V$ that is continuous in the topology of Corollary 2 and commutes with every X_i is a multiple of the identity.

This is a form of the condition Roberts imposes on labelled Hilbert spaces: "We simply suppose that $\mathcal{O}' = \mathcal{O}'_l$, where $\mathcal{O}'$ denotes the set of all bounded operators commuting with $\mathcal{O}$" [5, Sec. 2, p. 1099], where $\mathcal{O}$ is the set of all observables and $\mathcal{O}_l$ that of the labelled ones. Without superselection rules $\mathcal{O}' = \mathbb{C}1$, and the condition says that the labelled observables have only the scalars in their commutant. Definition 2.2 states this on V itself, with continuous instead of bounded operators, and so needs no Hilbert completion. For unbounded operators "commutant" can be made precise in several inequivalent ways; the statements below are for this one. A sufficient condition for continuity is easy to check: if T commutes with every X_i, it commutes with every $A \in \mathcal{A}$, and $\|Tv\| \leq C\|v\|$ for all v gives $p_A(Tv) = \|TAv\| \leq C\,p_A(v)$.

Theorem 2.3. *Let $\hbar \neq 0$.*

1. *Example 4 of the brief, $\mathbb{C}^2$ with the spin observables $S_i = \frac{\hbar}{2}\sigma_i$, is irreducible.*
2. *$\mathbb{C}^2 \otimes \mathbb{C}^2$ with the defining observables $S_i \otimes 1$ is a quantum vector space, with the same commutation relations, but it is not irreducible: $1 \otimes \sigma_3$ commutes with every $S_i \otimes 1$.*
3. *$\mathcal{S}(\mathbb{R})$ with position alone is not irreducible: multiplication by e^{ix} commutes with position.*
4. *Example 3 of the brief, $\mathcal{S}(\mathbb{R})$ with X, $P = -i\hbar\,d/dx$ and I, is irreducible.*

Proof. (1) A 2×2 matrix that commutes with σ_3 is diagonal, and a diagonal one that commutes with σ_1 is scalar. (2) The commutation relations and symmetry pass from S_i to $S_i \otimes 1$, and the remaining conditions hold in every finite dimension; finite dimension also makes $1 \otimes \sigma_3$ continuous. (3) Multiplication by e^{ix} is an isometry for the L^2 norm, hence continuous by the remark above, and it is not scalar.

(4) The ladder operators $a_\mp = X \pm d/dx$ lie in $\mathcal{A}$, since $d/dx = (i/\hbar)P$. On the Hermite functions ψ_n they act by $a_-\psi_n = \sqrt{2n}\,\psi_{n-1}$ and $a_+\psi_n = \sqrt{2(n+1)}\,\psi_{n+1}$. Write $c_n(f) = \langle \psi_n, f\rangle$ and $S_m f = \sum_{n<m} c_n(f)\psi_n$. The partial sums converge to f in $\mathcal{S}(\mathbb{R})$, because the coefficients of a Schwartz function decrease rapidly [8]. Since X is symmetric and d/dx antisymmetric, $c_n(a_- g) = \langle a_+\psi_n, g\rangle = \sqrt{2(n+1)}\,c_{n+1}(g)$; so $a_- g = 0$ forces $c_n(g) = 0$ for $n \geq 1$, and $g = c_0(g)\psi_0$. Now let T be continuous and commute with X and P. It commutes with $a_\mp$, so $a_- T\psi_0 = T a_-\psi_0 = 0$ and $T\psi_0 = z\psi_0$ for some z. Applying a_+ gives $T\psi_n = z\psi_n$ for every n by induction, hence $TS_m f = zS_m f$, and in the limit $Tf = zf$. □

Parts (1) and (2) also bear on the brief's claim, in the proof of its Theorem 5, that "two systems are physically equivalent if and only if their Lie algebras of observables are equivalent". The two spaces have isomorphic Lie algebras of defining observables and the same structure constants, but they are not isomorphic, as their dimensions differ. Irreducibility alone does not restore the claim: spin $\frac{1}{2}$ and spin 1 are both irreducible, with the same commutation relations. The system is determined by the representation, not by the Lie algebra; for the Heisenberg algebra the Stone–von Neumann theorem makes the irreducible representation unique once $\hbar$ is fixed.

2.3 The group form

A stronger version of the same repair is to make a group part of the data. Let G be a Lie group, π an irreducible unitary representation on a Hilbert space H, and let V be the space H^∞ of smooth vectors, with the seminorms $\|d\pi(u)v\|$ for u in the enveloping algebra. These are the seminorms of Corollary 2 for the defining observables $i\,d\pi(Y_k)$, with (Y_k) a basis of the Lie algebra. Antoine builds his space from such a representation and requires it to be invariant under the symmetry group [6]; he considers H^∞ and sets it aside because it need not be nuclear. For the Schrödinger representation of the Heisenberg group, H^∞ is $\mathcal{S}(\mathbb{R})$ with its usual topology [7, Secs. 1 and 6]. In this form every defining observable is essentially self-adjoint on V, as Antoine recalls from Segal and from Nelson and Stinespring, and it generates a unitary group that maps V onto itself. Every $\pi(g)$ maps V onto itself, which is the condition on changes of frame that Sec. 1 found missing from the brief's argument for [B]WDQ. The cost is that the definition presupposes the Hilbert space and the group. The Lean files contain only the Lie-algebra form above; Theorem 2.3(4) is its instance for the Heisenberg group.

2.4 Lean

Module:
`QuantumVectorSpaces.FurtherResults.Irreducibility`,
namespace `QuantumVectorSpaces.FurtherResults`.

- `not_completeSpace_posOnly`, `not_isQuantumVectorSpace_posOnly`: Theorem 2.1.
- `IsIrreducible`: Definition 2.2; `continuous_of_commute`: the sufficient condition for continuity.
- `isIrreducible_qubit`, `isQuantumVectorSpace_pair`, `not_isIrreducible_pair`, `not_isIrreducible_posOnly`, `isIrreducible_schwartz`: Theorem 2.3(1)–(4).
- The convergence of Hermite expansions in $\mathcal{S}(\mathbb{R})$ is `SchwartzMap.tendsto_hermiteSum`, in `ForMathlib.Analysis.Distribution.SchwartzSpace.HermitePartialSum`.

Not formalized: that Φ_X is a quantum vector space, the remark on Theorem 5, and the group form.

3 [B]WDQ relative to a group of frames

Section 1 and the companion note [20, Sec. 2] show that [B]WDQ cannot hold for "all frames" as the brief words it: some changes of frame do not map V into itself, and no nonzero state serves every smooth frame at once. This section states [B]WDQ with the frames as part of the data and shows that the brief's argument then goes through, with a condition that can be checked.

Definition 3.1. A *frame group* for V is a group F with a homomorphism ρ from F to the linear bijections of V such that every $\rho(g)$ preserves the inner product. The frame g has the defining observables $X_i^g = \rho(g)^{-1} X_i \rho(g)$: the observer in frame g describes the state v by $\rho(g)v$ and measures X_i on it. The frame group satisfies *[B]WDQ* if for all $g, h \in F$ the defining observables of g are observables of h, that is, symmetric operators continuous in the topology that Corollary 2 gives to the X_i^h.

The last condition is the premise of the brief's proof of Theorem 5, for every pair of frames in F. The frame 1 is the original one.

Theorem 3.2. *Let the X_i be symmetric and let (F, ρ) be a frame group.*

1. *(F, ρ) satisfies [B]WDQ if and only if every frame gives the same topology on V, and if and only if every $\rho(g)$ is continuous, that is, F acts by automorphisms of the topological vector space V.*

2. *Then all frames have the same observables, and every $\rho(g)$ is a unitary equivalence of V with itself. So it carries the pure states, with their topology, the Born probabilities and the observables of one frame to those of another.*

3. *A sufficient condition is that every X_i^g lies in the algebra $\mathcal{A}$ generated by the X_i.*

4. *The translations $f \mapsto f(\cdot - a)$, $a \in \mathbb{R}$, are a frame group of Example 3 that satisfies [B]WDQ, for every $\hbar$.*

5. *The change of frame $y = \tan(\frac{\pi}{2} \operatorname{erf} x)$ belongs to no frame group of Example 3: no linear map of $\mathcal{S}(\mathbb{R})$ agrees with its unitary U.*

Proof. (1) Since $\rho(g)$ preserves the inner product, the topology of the frame g is the topology of the frame 1 pulled back along $\rho(g)$; this is the transport result already used for isomorphisms (checked with the brief, `QuantumVectorSpaces.induced_topology_eq_pullback`). Conjugation by $\rho(g)$ also preserves symmetry. By the brief's frame criterion (`QuantumVectorSpaces.topology_eq_iff_continuous`), [B]WDQ for the pairs $(g, 1)$ and $(1, g)$ says that the topology of the frame g equals that of the frame 1, and conversely equal topologies give [B]WDQ for every pair. A topology t equals its pullback along $\rho(g)$ for every g exactly when every $\rho(g)$ is continuous for t: continuity of $\rho(g)$ gives one inclusion, and continuity of $\rho(g^{-1}) = \rho(g)^{-1}$ gives the other. (2) follows from (1). (3) If every X_i^g lies in $\mathcal{A}$, then $X_i = \rho(g)^{-1} X_i^{g^{-1}} \rho(g)$ lies in the algebra of the frame g, and two frames whose defining observables lie in each other's algebras give the same topology. (4) Under translation by a, position becomes $X + a$ and momentum stays P, both in $\mathcal{A}$; the inner product is preserved because Lebesgue measure is translation invariant. (5) is Theorem 1.1(1). □

So the brief's argument for [B]WDQ is correct once "all frames" is read as "all frames of a given frame group that acts by automorphisms of V". Part (1) says that this restriction is also necessary. Antoine requires the same invariance of his space under the symmetry group [6, Sec. II], and his Proposition 4, on observables that transform by a finite-dimensional representation, is related to part (3). The negative half is not in his paper: part (5), and the companion note's result that no space works for all smooth frames. Which frame group a quantum vector space should carry is a choice made with the space. For $\mathcal{S}(\mathbb{R})$ the changes of variable that preserve it are those of Galbis and Jordá [4, Thm. 2.3], with the half-density factor added for unitarity; in the group form of Sec. 2, F is the group G itself.

3.1 Lean

Module:
`QuantumVectorSpaces.FurtherResults.FrameGroups`,
namespace `QuantumVectorSpaces.FurtherResults`.

- `FrameGroup`, `frameObs`, `IsFrameCompatible`: Definition 3.1.
- `isFrameCompatible_iff_topology_eq`, `isFrameCompatible_iff_continuous`: Theorem 3.2(1).
- `IsFrameCompatible.isObservable_iff`, `IsFrameCompatible.unitaryEquiv`: Theorem 3.2(2); the pure states and observables are carried by the brief's `UnitaryEquiv.pureStates` and `UnitaryEquiv.isObservable_conj_iff`.
- `isFrameCompatible_of_mem_algebra`: Theorem 3.2(3).
- `translate`, `frameObs_translate`, `isFrameCompatible_translate`: Theorem 3.2(4).
- `not_exists_frameUnitary`: Theorem 3.2(5).

4 Ensembles over Example 3 and the continuity of entropy

QSTO:V speaks of density operators and of the von Neumann entropy $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 of the brief calls the set of ensembles "the most natural space to consider", but the brief defines no ensemble space over V. For Example 3 this section proposes one and shows that the entropy is finite and continuous on it. This is the quantum counterpart of the brief *Entropy is not continuous in standard probability spaces* [9]: there the entropy fails to be continuous in the L^1 topology, and here a topology made from the observables of V restores continuity.

Take $\hbar = 1$, so that $N = \frac{1}{2}(X^2 + P^2) - \frac{1}{2}$ has the Hermite functions ψ_n as eigenvectors, $N\psi_n = n\psi_n$. (For other $\hbar$ the Hermite functions are rescaled.) For a positive trace-class operator ρ on $L^2(\mathbb{R})$ write $\operatorname{tr} \rho N^j = \sum_n n^j \langle \psi_n, \rho\, \psi_n \rangle \in [0, \infty]$.

Definition 4.1. A *Schwartz density operator* is a positive operator ρ on $L^2(\mathbb{R})$ with $\operatorname{tr} \rho = 1$ and $\operatorname{tr} \rho N^j < \infty$ for every j. The ensembles over Example 3 are the Schwartz density operators, with the topology of Schwartz operators of Keyl, Kiukas and Werner [10].

The definition agrees with theirs. They show that the seminorms $\|H^m T H^{m'}\|_2$, $m, m' \in \mathbb{N}$, with $H = N + \frac{1}{2}$, induce the topology of Schwartz operators [10, Sec. 3.3]. For positive T, the operators $H^m T H^{m'}$ are products of the Hilbert–Schmidt operators $H^m T^{1/2}$ and $T^{1/2} H^{m'}$ exactly when every $\operatorname{tr}(T H^{2m}) = \|T^{1/2} H^m\|_2^2$ is finite, that is when all moments $\operatorname{tr} T N^j$ are finite. The eigenvectors of a Schwartz operator with nonzero eigenvalue are Schwartz functions [10, Lemma 3.5(c)], so a Schwartz density operator is a mixture of pure states of V. The expectation of every polynomial in X and P is continuous on it, since $|\operatorname{tr}(A\rho) - \operatorname{tr}(A\sigma)| \le \|A(\rho - \sigma)\|_1$ and the right-hand side is a seminorm of the topology [10, eq. (20)]. In particular convergence of Schwartz operators implies convergence in the trace norm and of $\operatorname{tr} \rho N$.

Theorem 4.2. *1. For every Schwartz function f with $\|f\| = 1$, the pure state $|f\rangle\langle f|$ is a Schwartz density operator.*

2. The entropy of a Schwartz density operator is finite.

3. (Wehrl.) *Let* $H = \sum_k \omega_k |b_k\rangle\langle b_k| \geq 0$ *on a Hilbert space, with* $\operatorname{tr} e^{-\beta H} < \infty$ *for all* $\beta > 0$. *If density operators* ρ_m *converge to* σ *in the trace norm and* $\operatorname{tr} \rho_m H \leq h$ *for all* m, *then* $S(\rho_m) \to S(\sigma) < \infty$.

4. *Hence the entropy is continuous on the ensembles over Example 3: if* $\|\rho_m - \sigma\|_1 \to 0$ *and* $\operatorname{tr} \rho_m N \to \operatorname{tr} \sigma N < \infty$, *then* $S(\rho_m) \to S(\sigma)$.

Part (3) is Wehrl's [11, Sec. II.D, p. 241]; that the condition on H is also necessary is Shirokov's [12, Prop. 1a]. Wehrl derives it from the lower semicontinuity of the entropy and of the free energy. The proof below needs neither, nor the ordering of eigenvalues, so that it could be checked in Lean with the trace-class theory at hand.

Proof. (1) The Hermite coefficients $c_n = \langle \psi_n, f \rangle$ of a Schwartz function decrease rapidly [8]: $(n+1)^{j+1}|c_n| \leq C$. So $\operatorname{tr}(|f\rangle\langle f| N^j) = \sum_n n^j |c_n|^2 \leq C^2 \sum_n (n+1)^{-2}$.

(3) Write $\rho = \sum_i \lambda_i |e_i\rangle\langle e_i|$ and $\eta(\lambda) = -\lambda \log \lambda$, so that $S(\rho) = \sum_i \eta(\lambda_i)$. For $K \in \mathbb{N}$ let $g_K(\lambda) = \min(-\log \lambda, K)$, continuous on $[0,1]$, and split $\eta(\lambda) = \lambda g_K(\lambda) + r_K(\lambda)$ with $r_K \geq 0$.

The head is continuous. For every $n \geq 1$, $\rho^n - \sigma^n = \rho(\rho^{n-1} - \sigma^{n-1}) + (\rho - \sigma)\sigma^{n-1}$, so by induction $\|\rho^n - \sigma^n\|_1 \leq n \|\rho - \sigma\|_1$ for density operators, and $|\operatorname{tr} \rho^n - \operatorname{tr} \sigma^n| \leq n \|\rho - \sigma\|_1$. Given $\varepsilon > 0$, choose by Weierstrass a polynomial q with $|g_K - q| \leq \varepsilon$ on $[0,1]$. Then $\sum_i \lambda_i q(\lambda_i) = \sum_j q_j \operatorname{tr} \rho^{j+1}$ is continuous in ρ, and it is within $\varepsilon \sum_i \lambda_i = \varepsilon$ of $\sum_i \lambda_i g_K(\lambda_i)$. So $\rho \mapsto \sum_i \lambda_i g_K(\lambda_i)$ is a uniform limit of continuous functions.

The tail is uniformly small. Klein's inequality $-a \log a \leq -a \log b + b - a$ with $b = e^{-K-1-\beta\omega_k}$ gives $r_K(\lambda) \leq \beta\omega_k \lambda + e^{-K-1} e^{-\beta\omega_k}$ for every k (for $\lambda \geq e^{-K}$ the left side is zero). Average over k with the weights $|\langle b_k, e_i \rangle|^2$, which sum to one in k and in i:

$$\sum_i r_K(\lambda_i) \leq \beta \operatorname{tr} \rho H + e^{-K-1} \operatorname{tr} e^{-\beta H}.$$

On $\{\operatorname{tr} \rho H \leq h\}$, choose β with $\beta h < \varepsilon/4$, then K with $e^{-K-1} \operatorname{tr} e^{-\beta H} < \varepsilon/4$. The tails of ρ_m and of σ lie in $[0, \varepsilon/2)$, since the energy is lower semicontinuous and so $\operatorname{tr} \sigma H \leq h$. Together with the head, $|S(\rho_m) - S(\sigma)| < \varepsilon$ eventually. The same bound with $K = 0$ gives $S(\rho) < \infty$ whenever $\operatorname{tr} \rho H < \infty$, and so (2), since $\operatorname{tr} e^{-\beta N} = (1 - e^{-\beta})^{-1}$.

(4) If $\operatorname{tr} \rho_m N \to \operatorname{tr} \sigma N$, the energies are eventually bounded, and (3) applies with $H = N$. □

The topology of Definition 4.1 is stronger than needed: by (4) the entropy is already continuous for the trace norm together with $\operatorname{tr} \rho N$, and on each set $\{\operatorname{tr} \rho N \leq h\}$ for the trace norm alone. Whether the ensembles induce on the pure states the quotient topology of V, which the brief needs for its remark on second countability, is not settled here. Woronowicz's description of the positive functionals on the operators of $\mathcal{S}(\mathbb{R})$ as the positive ρ with $\rho L^2 \subset \mathcal{S}$ [13] suggests that the class of Definition 4.1 is the natural one; the comparison was not carried out.

4.1 Lean

The general theorem is in `ForPhyslib.QuantumMechanics.EntropyContinuity`, namespace `QuantumMechanics`; the application in `QuantumVectorSpaces.FurtherResults.Ensembles`, namespace `QuantumVectorSpaces.FurtherResults`.

- `numberMoment`, `IsSchwartzDensity`: Definition 4.1, without the topology; the Hermite basis is TauCeti's `hermiteHilbertBasis`.

- `isSchwartzDensity_proj`: Theorem 4.2(1).
- `IsSchwartzDensity.vonNeumannEntropy_ne_top`, from `vonNeumannEntropy_ne_top`: Theorem 4.2(2).
- `tendsto_vonNeumannEntropy`: Theorem 4.2(3), with the energy bound required only eventually; `traceENorm_pow_sub_pow_le`, `tendsto_eigenSum` (the head), `tailSum_le` (the tail), `energy_le_of_tendsto` (lower semicontinuity of the energy).
- `tendsto_vonNeumannEntropy_of_tendsto_numberMoment`: Theorem 4.2(4).

Not formalized: the topology of Schwartz operators and the comparison with Keyl, Kiukas and Werner's definition; the continuity of the expectations of polynomials in X and P.

5 A space of states that remembers the number of degrees of freedom

[B]STDIST asks that systems with different numbers of degrees of freedom be distinct. QSTO:V gets this from the defining observables (proof of Theorem 5), not from V: as topological vector spaces the Schwartz spaces are all isomorphic. The brief *Unphysicality of Hilbert spaces* [2] says that "Schwartz spaces present the same problem", and its formalization proves this: for all $n, m \geq 1$ a unitary $L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^m)$ maps $\mathcal{S}(\mathbb{R}^n)$ onto $\mathcal{S}(\mathbb{R}^m)$ as an isomorphism of topological vector spaces (`SchwartzMap.exists_unitary_schwartzEquiv`). This section shows that a slightly smaller space does remember n, and what that space costs in Definition 1.

For $\mu \geq \frac{1}{2}$ the Gelfand–Shilov space $S^\mu_\mu(\mathbb{R}^n)$ consists of the smooth u with $|x^\beta \partial^\alpha u(x)| \leq CA^{|\alpha|+|\beta|}(\alpha!\,\beta!)^\mu$ for some constants A, C. It carries the inductive limit topology, as $A \to \infty$, of the Banach spaces in which the least such C is the norm. It contains the Hermite functions h_k, and $S^{1/2}_{1/2}$ is the smallest nonzero space of the scale [15, Sec. 1]. Its elements are the functions whose Hermite coefficients decrease exponentially [14, Remark 1.3], that is, the analytic vectors $\bigcup_{t>0} D(e^{tN})$ of the number operator N.

Theorem 5.1. *Let $\mu, \nu \geq \frac{1}{2}$ and $n, m \geq 1$. Then $S^\mu_\mu(\mathbb{R}^n)$ and $S^\nu_\nu(\mathbb{R}^m)$ are isomorphic as locally convex spaces if and only if $\mu n = \nu m$. In particular $S^{1/2}_{1/2}(\mathbb{R}^n) \cong S^{1/2}_{1/2}(\mathbb{R}^m)$ only for $n = m$.*

For $\mu, \nu > \frac{1}{2}$ this follows from Langenbruch's description of the coefficient spaces [16, Example 5.3(b)] together with Vogt's classification of power series spaces [17, Thm. 5.12]. Langenbruch's introduction claims $r \geq \frac{1}{2}$, but his Remark 5.2 and Example 5.3(b) need a logarithmic factor in the defining sequence at $r = \frac{1}{2}$. So the case that matters here is not covered by a formal statement there. The proof below treats all $\mu \geq \frac{1}{2}$ alike.

Proof. For an exponent sequence $0 \leq a_0 \leq a_1 \leq \cdots \to \infty$, the power series space of finite type is

$$\Lambda_0(a) = \Big\{ x : |x|_t^2 = \sum_j |x_j|^2 e^{2ta_j} < \infty \text{ for all } t < 0 \Big\},$$

a Fréchet–Schwartz space [17, Thm. 5.8].

Coefficients. Enumerate the multi-indices $k \in \mathbb{N}^n$ as $k(0), k(1), \ldots$ with $|k(j)|$ nondecreasing. There are $\binom{M+n}{n}$ multi-indices with $|k| \leq M$, a number between $M^n/n!$ and $(M+1)^n$. Hence $\frac{1}{2} j^{1/n} \leq |k(j)| \leq (n!\,j)^{1/n}$ for $j \geq 1$. Gramchev, Pilipović and Rodino [14, Thm. 1.2] show that $f \in L^2(\mathbb{R}^n)$

lies in $S^\mu_\mu(\mathbb{R}^n)$ if and only if its Hermite coefficients $c_j = \langle h_{k(j)}, f\rangle$ satisfy $\sum_j |c_j|^2 e^{2sj^{1/(2n\mu)}} < \infty$ for some $s > 0$. (Their eigenvalues are those of the harmonic oscillator, $2|k|+n$, in this order.) Lozanov-Crvenković and Perišić [15, Thm. 2.3] show that the coefficient map is a topological isomorphism onto this sequence space. The sequence space carries the inductive limit topology of the weighted ℓ^2 spaces as $s \to 0$. Their weights are built from the $M(\theta_i\sqrt{k_i})$, $\theta_i > 0$, with $M(\rho) \asymp \rho^{1/\mu}$ for large ρ. Since $|k|^{1/(2\mu)} \le \sum_i k_i^{1/(2\mu)} \le n|k|^{1/(2\mu)}$ and $|k(j)| \asymp j^{1/n}$, they define the same inductive limit. This inductive limit is the strong dual $\Lambda_0(a^{n,\mu})'_b$ with $a_j^{n,\mu} = j^{1/(2n\mu)}$. Indeed, the dual of $\Lambda_0(a)$ is the union of the duals $\ell^2(e^{-2ta_j})$ of its local Hilbert spaces. Since $\Lambda_0(a)$ is reflexive, hence distinguished [17, Thms. 3.3, 2.8], the strong topology is the inductive one [17, Sec. 2]. So $S^\mu_\mu(\mathbb{R}^n) \cong \Lambda_0(a^{n,\mu})'_b$.

Duality. An isomorphism between the strong duals of two reflexive Fréchet spaces transposes to an isomorphism between their strong biduals, which are the spaces themselves. So $S^\mu_\mu(\mathbb{R}^n) \cong S^\nu_\nu(\mathbb{R}^m)$ if and only if $\Lambda_0(a^{n,\mu}) \cong \Lambda_0(a^{m,\nu})$.

Classification. By [17, Thm. 5.12], $\Lambda_0(a) \cong \Lambda_0(b)$ if and only if $a_j/C \le b_j \le Ca_j$ for some C and all large j. For $a_j = j^{1/(2n\mu)}$ and $b_j = j^{1/(2m\nu)}$ this holds exactly when the exponents agree, that is when $n\mu = m\nu$. □

At $\mu = \frac{1}{2}$, where the literature was in doubt, the topological half of the coefficient description can also be checked without [15]. Both $S^{1/2}_{1/2}(\mathbb{R}^n)$ and the coefficient space are countable inductive limits of Banach spaces. By [14] the coefficient map is a bijection between them. The map and its inverse have closed graphs, because every coefficient $f \mapsto \langle h_k, f\rangle$ is continuous on both spaces. By Grothendieck's closed graph theorem for such spaces [18], both are continuous.

Why the Schwartz spaces do not remember n. The same argument for $\mathcal{S}(\mathbb{R}^n)$ gives a power series space of infinite type with exponents $\log j^{1/n} = \frac{1}{n}\log j$. Vogt's criterion allows a constant factor, so every $\mathcal{S}(\mathbb{R}^n)$ is isomorphic to $s = \Lambda_\infty(\log(j+1))$. For $S^{1/2}_{1/2}$ the dimension sits in the exponent of $j^{1/n}$, where no constant can absorb it. The isomorphism in the formalization of the Hilbert brief relabels Hermite functions along a bijection $\sigma\colon \mathbb{N}^n \to \mathbb{N}^m$ with $|\sigma(k)|$ polynomially bounded in $|k|$, and conversely. For $S^{1/2}_{1/2}$ a relabelling would need $|\sigma(k)| \asymp |k|$, which counting excludes when $n \neq m$. Theorem 5.1 shows that no isomorphism of any kind exists.

Only the product μn is remembered. For example $S^{1/2}_{1/2}(\mathbb{R}^2) \cong S^1_1(\mathbb{R})$. So the number of degrees of freedom becomes a topological invariant only once μ is fixed. The natural choice is $\mu = \frac{1}{2}$, the smallest nonzero space of the scale.

5.1 The cost in Definition 1

Theorem 5.2. *1. With its own topology, $V = S^{1/2}_{1/2}(\mathbb{R}^n)$ is not metrizable, hence not second countable. So no choice of defining observables whose topology of Corollary 2 is this one makes V a quantum vector space.*

2. The observables X, P, I of Example 3 map $S^{1/2}_{1/2}(\mathbb{R})$ into itself. With them, $S^{1/2}_{1/2}(\mathbb{R})$ is not complete in the topology of Corollary 2, so it is not a quantum vector space, whatever the structure constants.

Proof. (1) V is the union of the Hilbert spaces $H_k = \{f : \sum_\alpha |c_\alpha(f)|^2 e^{2|\alpha|/k} < \infty\}$, each continuously included. As the strong dual of a Fréchet space, V is complete [17, Cor. 2.6]. If it were metrizable it would be a Fréchet space, and by Baire's theorem some H_k would be of second category in V.

Banach's argument (as in the proof of the open mapping theorem, [17, Lemmas 1.4, 1.5]) would then make the inclusion $H_k \to V$ surjective. But the function with coefficients $e^{-|\alpha|/k}$ lies in H_{2k} and not in H_k. A topological vector space with a countable base at 0 is metrizable, so V is not second countable, which Definition 1 requires.

(2) Write $a_\mp = X \pm D$, so that $a_- h_j = \sqrt{2j}\, h_{j-1}$ and $a_+ h_j = \sqrt{2(j+1)}\, h_{j+1}$. Then $c_j(a_- f) = \sqrt{2(j+1)}\, c_{j+1}(f)$ and $c_j(a_+ f) = \sqrt{2j}\, c_{j-1}(f)$, and these coefficients still decrease exponentially, with half the rate. As X and D are combinations of $a_\pm$, the space is invariant. On an invariant subspace, the topology of Corollary 2 for the restricted observables is the subspace topology. For Example 3 that is the topology of $\mathcal{S}(\mathbb{R})$ (checked with the brief). The space contains the Hermite partial sums of every Schwartz function, which converge in $\mathcal{S}(\mathbb{R})$ [8], so it is dense. The Schwartz function with coefficients $e^{-\sqrt{j+1}}$ is not in it. A dense proper subspace is not closed, hence not complete. □

So $S^{1/2}_{1/2}$ is not a quantum vector space as Definition 1 stands. It needs its own inductive limit topology, which no family of defining observables can give it under second countability. The observables of Example 3 give it the wrong topology. Using it means changing Definition 1 to allow such topologies. That gives up second countability, which the brief ties to the second countability of the pure states (Sec. 4).

5.2 Moments determine the pure states

In $\mathcal{S}(\mathbb{R})$ two different density operators can have the same moments of X and P. This is Woronowicz's Counterexample II [19], and the construction of the brief *Non-uniqueness of the moment problem for Schwartz functions*. It uses functions with disjoint compact supports. $S^{1/2}_{1/2}$ is a quasi-analytic class [15, Sec. 1] and has no nonzero functions of compact support, and the non-uniqueness disappears.

Theorem 5.3. *Take $\hbar = 1$. Let $\psi \in S^{1/2}_{1/2}(\mathbb{R})$ with $\|\psi\| = 1$. If $\rho \geq 0$ with $\rho L^2(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$ satisfies $\operatorname{tr}(\rho A) = \langle \psi, A\psi \rangle$ for every polynomial A in X and P, then $\rho = |\psi\rangle\langle\psi|$.*

Proof. Woronowicz [19, Theorem] proves the following. If a state ω on the polynomials in X and P satisfies $\sum_k \omega(a^k (a^\dagger)^k)^{1/2}\, t^k/k! < \infty$ for some $t > 0$, with $a^\dagger h_j = \sqrt{j+1}\, h_{j+1}$, then exactly one positive ρ with $\rho L^2 \subset \mathcal{S}$ represents it. For $\omega = \langle \psi, \cdot\, \psi \rangle$ and $|c_j(\psi)| \leq C e^{-\varepsilon j}$,

$$\omega(a^k (a^\dagger)^k) = \|(a^\dagger)^k \psi\|^2 = \sum_j |c_j|^2 \frac{(j+k)!}{j!} = k! \sum_j |c_j|^2 \binom{j+k}{j}.$$

The binomial theorem gives $\binom{j+k}{j} p^j (1-p)^k \leq 1$ for $0 < p < 1$. With $p = \delta/(1+\delta)$ this reads $\binom{j+k}{j} \leq (1+\delta)^k (1+1/\delta)^j$. Choose δ with $1 + 1/\delta < e^{2\varepsilon}$, so that $S = \sum_j |c_j|^2 (1+1/\delta)^j < \infty$. Then $\|(a^\dagger)^k \psi\| \leq \sqrt{k!\, S}\, (1+\delta)^{k/2}$, and the series is at most $\sqrt{S} \sum_k (t\sqrt{1+\delta})^k / \sqrt{k!} < \infty$ for every t. □

5.3 Lean

Module:
`QuantumVectorSpaces.FurtherResults.GelfandShilov`,
namespace `QuantumVectorSpaces.FurtherResults`. There $S^{1/2}_{1/2}(\mathbb{R})$ is defined by its Hermite coefficients [14, Remark 1.3], not by the Gelfand–Shilov estimates.

- `ExpDecay`, `gelfandShilov`: the Schwartz functions with $|c_j| \leq C e^{-\varepsilon j}$, as a subspace of $\mathcal{S}(\mathbb{R})$.

- `coreRestrict`, `obsRestrict`, `topology_restrict`: on a subspace invariant under the defining observables, the topology of Corollary 2 is the subspace topology (any V, any observables).
- `coef_raiseOp`, `lowerOp_apply_mem`, `raiseOp_apply_mem`, `obs_apply_mem`: the ladder operators, and so X, P, I, preserve the space.
- `topology_gelfandShilov`, `dense_gelfandShilov`, `exists_not_mem_gelfandShilov`, `not_completeSpace_gelfandShilov`, `not_isQuantumVectorSpace_gelfandShilov`: Theorem 5.2(2).

Not formalized: Theorem 5.1, which needs the Gelfand–Shilov estimates for Hermite expansions, inductive limit topologies and the classification of power series spaces; and Theorems 5.2(1) and 5.3.

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